CAN–Ethernet Bridge IP Core

Functional Specification

Bidirectional CAN 2.0B ↔ 100 Mbps Ethernet Gateway/Tunnel — Interface Definitions, Functional Requirements, Register Map, Protocol Specification

IIITDM Chennai · Final Year Project · Rev 1.0 · May 2026

Contents

1. Scope and Purpose

This document specifies the functional behaviour of the CAN–Ethernet Bridge IP core. It covers interface definitions, the custom encapsulation protocol, register specifications, functional requirements, and timing constraints.

The bridge enables transparent, bidirectional data transfer between a CAN 2.0B bus (1 Mbps) and a 100 Mbps Ethernet network. It operates as a Wishbone-based IP core with three bus interfaces: an 8-bit WB master to the CAN controller, a 32-bit WB master to the Ethernet MAC/RAM, and a 32-bit WB slave for host CPU configuration.

2. System Overview

Parameter	Value	Notes
CAN interface	8-bit Wishbone master	SJA1000-compatible register set
ETH interface	32-bit Wishbone master	OpenCores Ethernet MAC + shared RAM
Host interface	32-bit Wishbone slave	32 registers (0x00–0x7C)
Operating modes	Gateway / Tunnel	Software-selectable
CAN ID filter	16 entries, 29-bit	Accept-list or reject-list
CAN TX queue	8-deep FIFO	Absorbs ETH→CAN speed gap
Clock domain	Single (50 MHz)	No internal CDC
Reset	Active-low synchronous	Hybrid defaults (CPU-less boot)

3. Top-Level Interface

Module: `can_eth_bridge_top` (`rtl/can_eth_bridge_top.v`)

3.1 Clock and Reset

Port	Dir	Width	Description
`clk`	in	1	System clock. All registers are positive-edge triggered. Nominal 50 MHz.
`rst_n`	in	1	Active-low synchronous reset. All FSMs return to IDLE; config registers load hybrid defaults.

3.2 CAN Wishbone Master (8-bit)

Port	Dir	Width	Description
can_wbm_adr_o	out	8	CAN controller register address
can_wbm_dat_o	out	8	Write data to CAN controller
can_wbm_dat_i	in	8	Read data from CAN controller
can_wbm_cyc_o	out	1	Bus cycle active
can_wbm_stb_o	out	1	Valid transfer strobe
can_wbm_we_o	out	1	Write enable
can_wbm_ack_i	in	1	Slave acknowledge

3.3 ETH Wishbone Master (32-bit)

Port	Dir	Width	Description
eth_wbm_adr_o	out	32	Memory / BD address
eth_wbm_dat_o	out	32	Write data
eth_wbm_dat_i	in	32	Read data
eth_wbm_sel_o	out	4	Byte lane select
eth_wbm_cyc_o	out	1	Bus cycle active
eth_wbm_stb_o	out	1	Valid transfer strobe
eth_wbm_we_o	out	1	Write enable
eth_wbm_ack_i	in	1	Slave acknowledge

3.4 Host Wishbone Slave (32-bit)

Port	Dir	Width	Description
host_wb_adr_i	in	8	Register address (0x00–0x7C)
host_wb_dat_i	in	32	Write data from host
host_wb_dat_o	out	32	Read data to host
host_wb_cyc_i	in	1	Bus cycle
host_wb_stb_i	in	1	Strobe
host_wb_we_i	in	1	Write enable
host_wb_ack_o	out	1	Acknowledge

3.5 Interrupts

Port	Dir	Width	Description
can_irq_i	in	1	CAN frame received (triggers upstream FSM)
eth_tx_irq_i	in	1	Ethernet TX complete (upstream release)
eth_rx_irq_i	in	1	Ethernet frame received (triggers downstream FSM)

Port	Dir	Width	Description
`can_tx_irq_i`	in	1	CAN TX complete (downstream free BD)

4. Register Map

All registers are 32-bit, accessed at 4-byte aligned offsets via the Host WB slave.

Offset	Register	Access	Reset	Bit-fields
0x00	BRIDGE_CTRL	R/W	0x00000001	[0] bridge_en, [2:1] mode (01=GW, 10=TUN)
0x04	BRIDGE_STATUS	RO	0x00000000	[0] up_busy, [1] dn_busy, [7:4] queue_depth
0x08	DST_MAC_HI	R/W	0xFFFFFFFF	Destination MAC [47:16]
0x0C	DST_MAC_LO	R/W	0x0000FFFF	Destination MAC [15:0]
0x10	SRC_MAC_HI	R/W	0x02CAFE00	Source MAC [47:16]
0x14	SRC_MAC_LO	R/W	0x00000001	Source MAC [15:0]
0x18	FILTER_CTRL	R/W	0x00000000	[0] filter_en, [1] filter_mode
0x1C–0x58	FILTER_TBL[0:15]	R/W	0x00000000	16 × 29-bit CAN ID filter entries
0x5C	CNT_CAN_RX	RO	0x00000000	CAN frames received
0x60	CNT_ETH_TX	RO	0x00000000	Ethernet frames transmitted
0x64	CNT_ETH_RX	RO	0x00000000	Ethernet frames received
0x68	CNT_CAN_TX	RO	0x00000000	CAN frames transmitted
0x6C	CNT_FILTERED	RO	0x00000000	Frames dropped by filter
0x70	CNT_ERRORS	RO	0x00000000	Validation errors
0x74	TUNNEL_SEQ	RO	0x00000000	Tunnel sequence counter
0x78	TX_BD_ADDR	R/W	0x00000400	ETH TX buffer descriptor address
0x7C	RX_BD_ADDR	R/W	0x00000600	ETH RX buffer descriptor address

5. Encapsulation Protocol

5.1 Frame Format (Word-level, 32-bit aligned)

Word	Bits	Field
0	[31:0]	DST MAC [47:16]
1	[31:16]	DST MAC [15:0]
1	[15:0]	SRC MAC [47:32]
2	[31:0]	SRC MAC [31:0]
3	[31:16]	EtherType (`0xCAFE` gateway, `0xCABE` tunnel)

Word	Bits	Field
3	[15:8]	Magic high byte (`0xB1`)
3	[7:0]	Magic low byte (`0xDE`)
4	[31:24]	Protocol version (`0x01`)
4	[23:0]	CAN ID upper bits
5	[31:28]	CAN ID lower / EFF / RTR flags
5	[27:24]	DLC (0–8)
5	[23:16]	Flags (mode, filter status)
5	[15:0]	Reserved
6	[31:16]	Tunnel sequence number (tunnel mode only)
6	[15:0]	Timestamp (20 μs resolution)
7–8	[31:0]	CAN payload data (up to 8 bytes, zero-padded)
9–15	[31:0]	Padding to 64-byte Ethernet minimum

5.2 Downstream Validation Rules

A received Ethernet frame is accepted if **all** of the following are true:

1. EtherType matches the current bridge mode (`0xCAFE` for gateway, `0xCABE` for tunnel)
2. Magic bytes = `0xB1DE`
3. Protocol version = `0x01`
4. DLC ≤ 8

If any check fails, the frame is dropped and `cnt_errors` is incremented.

6. Functional Requirements

6.1 Operating Modes

ID	Requirement
FR-01	The bridge shall support **gateway mode** (EtherType `0xCAFE`).
FR-02	The bridge shall support **tunnel mode** (EtherType `0xCABE`) with auto-incrementing sequence numbers and timestamps.
FR-03	The bridge shall be software-disablable via `bridge_en` bit; when disabled, no frames shall be forwarded.
FR-04	Mode switching shall take effect within 1 clock cycle of register write.

6.2 Upstream Path (CAN → ETH)

ID	Requirement
FR-10	Upon CAN RX interrupt, the bridge shall read 13 bytes from the CAN controller (frame info + up to 8 data bytes).
FR-11	Standard (11-bit) and Extended (29-bit, EFF) CAN IDs shall be supported.
FR-12	DLC values 0–8 shall be accepted; DLC $>$ 8 shall be treated as 8.
FR-13	If the CAN ID filter is enabled, frames matching the filter criteria shall be dropped before encapsulation.
FR-14	The encapsulated Ethernet frame shall be written to shared RAM (16 $\times$ 32-bit words).
FR-15	After RAM write, the bridge shall program the TX Buffer Descriptor with length and READY bit.
FR-16	The bridge shall wait for ETH TX complete interrupt before releasing the buffer.
FR-17	Counter `cnt_can_rx` shall increment on each successful CAN read; `cnt_eth_tx` on each ETH transmit.

6.3 Downstream Path (ETH → CAN)

ID	Requirement
FR-20	Upon ETH RX interrupt, the bridge shall read the RX Buffer Descriptor (2 words).
FR-21	The bridge shall read the Ethernet frame from RAM (16 $\times$ 32-bit words).
FR-22	The frame shall be validated: EtherType, magic bytes (`0xB1DE`), protocol version (`0x01`), and DLC $\leq$ 8.
FR-23	Invalid frames shall be dropped; `cnt_errors` shall increment.
FR-24	Valid frames shall be decapsulated and the CAN frame enqueued in the TX queue.
FR-25	If the TX queue is full (8 entries), the FSM shall stall until space is available.
FR-26	The bridge shall write 13 bytes to the CAN controller and issue a TX request.
FR-27	The bridge shall wait for CAN TX complete interrupt, then free the RX BD.

6.4 CAN ID Filter

ID	Requirement
FR-30	The filter shall support 16 entries, each holding a 29-bit CAN ID.
FR-31	**Accept-list mode** (`filter_mode=0`): only IDs present in the table are forwarded.
FR-32	**Reject-list mode** (`filter_mode=1`): IDs present in the table are dropped.
FR-33	When `filter_en=0`, all frames shall pass (no filtering).
FR-34	`cnt_filtered` shall increment for each dropped frame.

6.5 Bus Arbitration

ID	Requirement
FR-40	The CAN bus and ETH bus shall each have a round-robin arbiter.
FR-41	Both upstream and downstream FSMs may request either bus simultaneously.
FR-42	No grant overlap: only one master shall be granted at any time per bus.
FR-43	No starvation: alternating priority ensures both FSMs are served.

6.6 Reset Behaviour

ID	Requirement
FR-50	On assertion of `rst_n=0`, all FSMs shall return to IDLE within 1 clock.
FR-51	No partial Buffer Descriptors shall be left in READY state after reset.
FR-52	All counters shall be cleared to zero on reset.
FR-53	Configuration registers shall load hybrid defaults (bridge enabled, gateway mode, broadcast DST MAC).

7. Timing Requirements

Parameter	Value	Notes
Target clock frequency	50 MHz	Single domain, no CDC
Upstream latency	$\sim$8.75 μs	CAN RX IRQ $\rightarrow$ BD READY
Downstream latency	$\sim$232 μs	ETH RX IRQ $\rightarrow$ CAN TX request
Timestamp resolution	20 μs	50 MHz / 1000 prescaler

Parameter	Value	Notes
CAN WB transfer	1 cycle/byte	8-bit bus, single-cycle ACK
ETH WB transfer	1 cycle/word	32-bit bus, single-cycle ACK
Max upstream throughput	∼114 kfps	CAN frames per second
Max downstream throughput	∼4.3 kfps	Bottleneck: 13-byte CAN writes

8. Error Handling

Error Condition	Bridge Response
Invalid EtherType	Frame dropped, `cnt_errors` incremented
Invalid magic (≠`0xB1DE`)	Frame dropped, `cnt_errors` incremented
Invalid version (≠`0x01`)	Frame dropped, `cnt_errors` incremented
DLC > 8	Frame dropped, `cnt_errors` incremented
CAN TX queue full	FSM stalls at QUEUE_CHK until space available
Mid-transaction reset	FSMs return to IDLE, no partial BD left READY

9. Revision History

Rev	Date	Author	Changes
1.0	May 2026	FYP / IIITDM	Initial release. Covers gateway/tunnel modes, full register map, encapsulation protocol, 28 functional requirements.